Pivoting the Simplex Tableau

row operations, pivots, and the simplex decision rules • *Linear Algebra*

Quiz Yourself

Cover the answers page. Say each answer out loud or write it down, then check yourself.

Card 1. What are the three elementary row operations you are allowed to perform on a matrix?

Card 2. In a matrix that has been fully reduced, which variables are called *basic* and which are called *free*?

Card 3. In a simplex tableau for a maximization problem, what rule tells you which variable enters the basis (that is, which column becomes the pivot column)?

Card 4. Once the pivot column is chosen in a simplex tableau, how do you decide which row to pivot in?

Card 5. After a pivot, the bottom (objective) row of your maximization tableau has no negative entries in the decision-variable columns. What does this result tell you, and what do you do next?

Answers

Card 1. (1) Switch two rows. (2) Multiply a row by a nonzero number. (3) Replace a row by itself plus a multiple of another row.

Card 2. Variables whose columns contain a pivot position are *basic* variables; variables whose columns contain no pivot position are *free* variables. The basic variables can be solved for in terms of the free ones.

Card 3. Look at the objective (bottom) row and choose the column with the *most negative* entry. That column's variable enters the basis, because increasing it improves the objective value fastest.

Card 4. Use the minimum-ratio test: for every row whose entry in the pivot column is positive, divide the right-hand-side value by that entry, and pivot in the row with the *smallest* ratio. For example, if the pivot column has entries 5 and 2 with right-hand sides 40 and 30, the ratios are $40/5 = 8$ and $30/2 = 15$, so you pivot in the first row.

Card 5. The tableau is *optimal*: no variable can enter the basis and increase the objective any further. You stop pivoting and read the maximum objective value from the right-hand side of the bottom row.